

Companion to Bird & de Moor, Algebra of Programming

§4. Relations and Allegories

§4.1. Allegories

§4.2. Special properties of arrows

$R:A \rightarrow B$, $S:B \rightarrow A$, $1_A \sqsubseteq RS$, $SR \sqsubseteq 1_B \implies S=R^\circ$ and R is a map

Proposition 4.1

A relation with a two-sided partial inverse under composition — identity below one order, identity above the other — is that inverse's converse, and R is itself a map.

§4.3. Tabular allegories

(f,g) a tabulation of $R \implies k^\circ h \sqsubseteq R \iff \exists! \text{ map } m \text{ with } h=mf, k=mg$

Proposition 4.2

Every pair below R 's tabulating pair factors, uniquely, through a single map into the tabulation's apex.

§4.4. Locally complete allegories

§4.5. Boolean allegories

§4.6. Power allegories